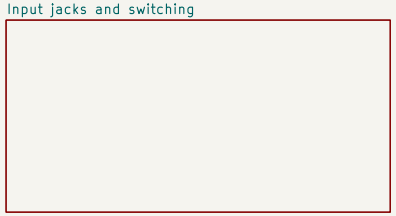
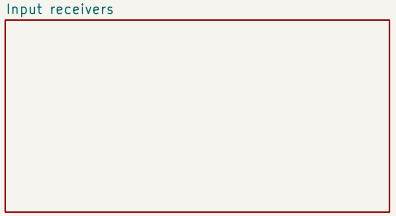
Plug & Roll main board. Each block of the board is one sub-sheet.

Input jacks and switching



File: input-jacks.kicad_sch

Input receivers



File: input.kicad_sch

INST buffer



File: input-inst.kicad_sch

Input gain and ADC pad



File: input-gain.kicad_sch

DAC gain and bypass



File: output-dac.kicad_sch

Sheet: /
File: main-board.kicad_sch

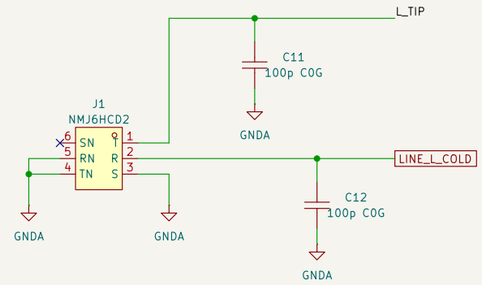
Title: Plug & Roll - Main board

Size: A4	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 1/6

INPUT JACKS

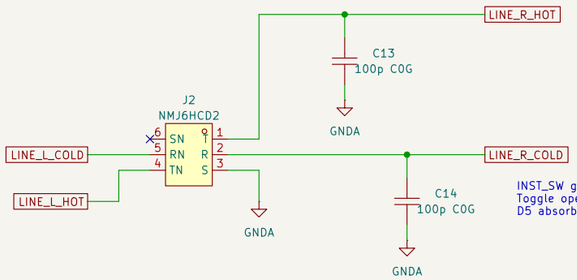
Tip = hot (+), ring = cold (−), sleeve = analog ground.
TN / RN / SN are switch contacts that rest on T / R / S while the jack is empty.

LEFT input



Empty jack: tip and ring are grounded (silent input).

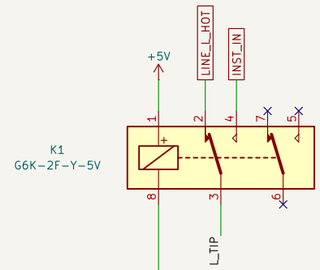
RIGHT input



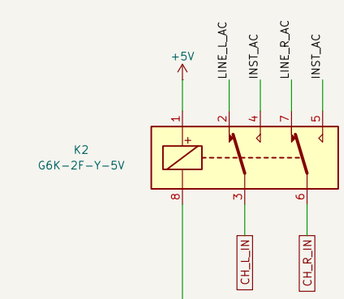
Empty jack: the Right receiver gets the Left signal (mono).
Purely analog, the Daisy is not involved.

INST / LINE SWITCHING

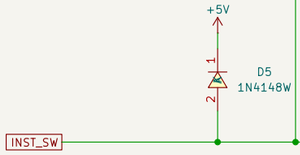
Relays drawn released (no coil current) = LINE. Panel toggle closed = coils on = INST.



K1: Left tip to the LINE receiver or to the INST buffer.
Second contact set not used.



K2: each channel takes its LINE receiver or the INST buffer.

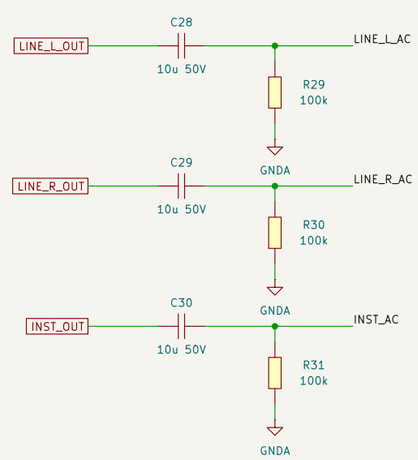


INST_SW goes through the ribbon to the INST/LINE toggle on the control board.
Toggle open = LINE. Toggle closed to ground = both coils on = INST.
D5 absorbs the voltage spike of the coils when the toggle opens.



5 V decoupling, placed at the relays

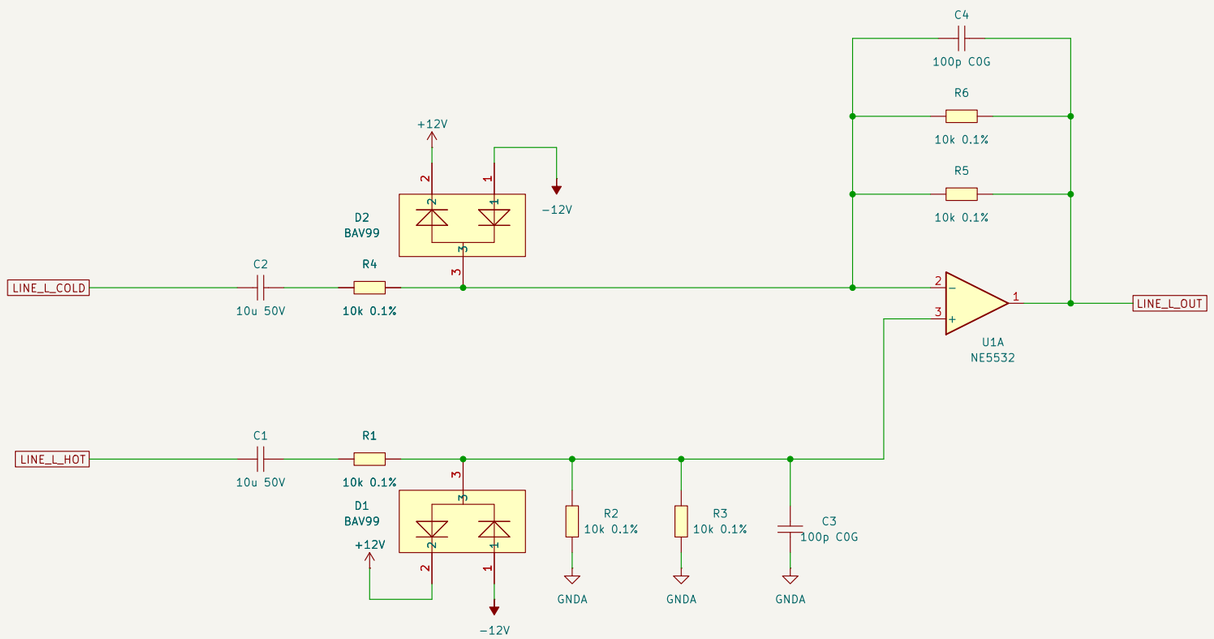
Coupling capacitors in front of K2



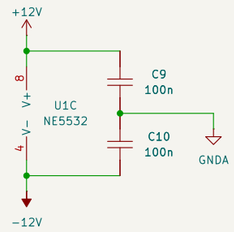
The three sources reach K2 without their DC offset,
so switching INST / LINE gives no DC step (click).

Sheet: /Input jacks and switching/		
File: input−jacks.kicad_sch		
Title: Plug & Roll − Main board − Input jacks and INST/LINE switching		
Size: A3	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 2/6

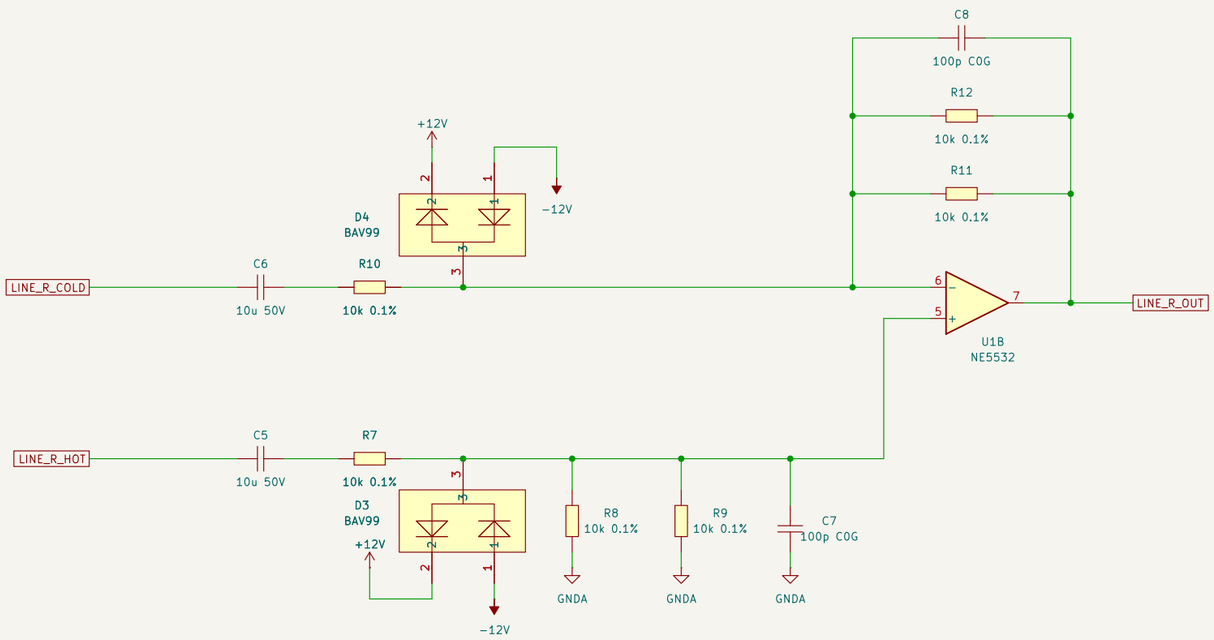
LINE receiver L: balanced input, gain x0.5 (−6 dB). All resistors 10k 0.1 %.



U1 supply and decoupling



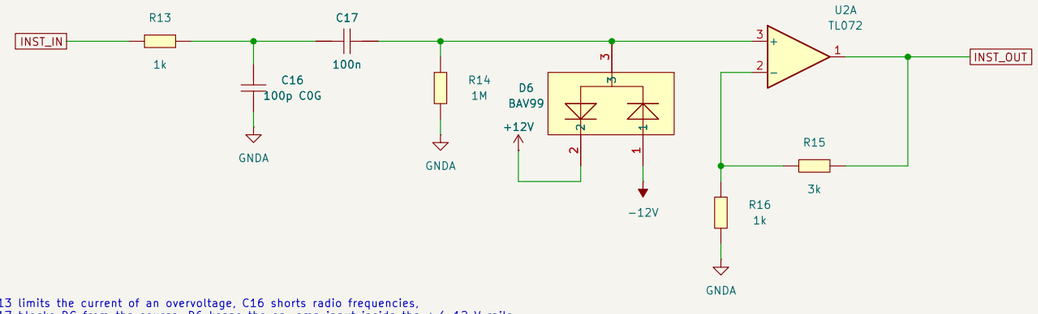
LINE receiver R: balanced input, gain x0.5 (−6 dB). All resistors 10k 0.1 %.



Sheet: /Input receivers/	
File: input.kicad_sch	
Title: Plug & Roll — Main board — Input receivers	
Size: A3	Date:
KiCad E.D.A. 10.0.6	**Rev: 0.1**
	Id: 3/6

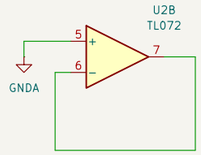
INST BUFFER

Instrument (guitar) input, used only in INST mode. Input resistance 1 Mohm, gain x4 (+12 dB) = 1 + R15/R16.

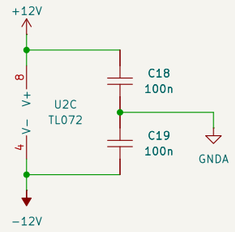


R13 limits the current of an overvoltage, C16 shorts radio frequencies,
C17 blocks DC from the source, D6 keeps the op-amp input inside the +/−12 V rails.

Unused half of U2



U2 supply and decoupling



Sheet: /INST buffer/
File: input-inst.kicad_sch

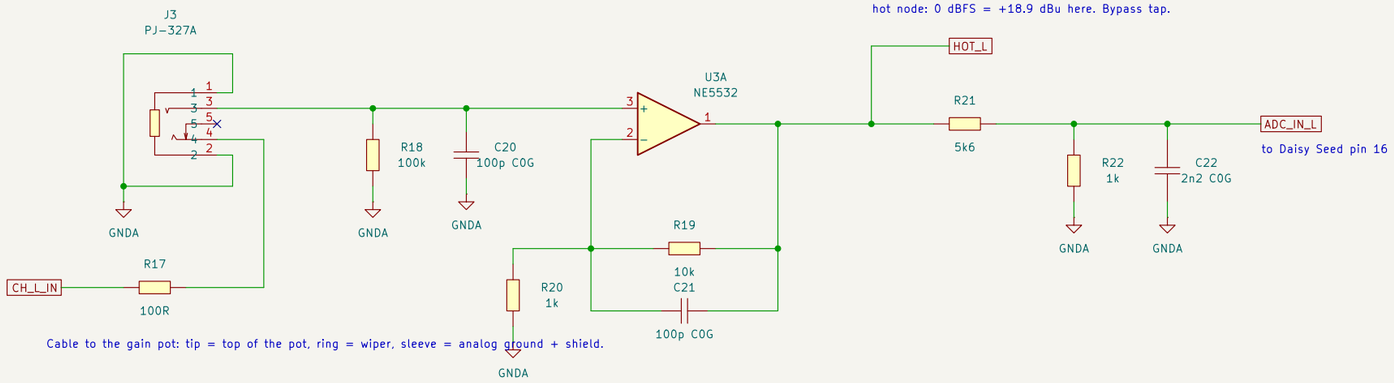
Title: Plug & Roll – Main board – INST buffer

Size: A3	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 4/6

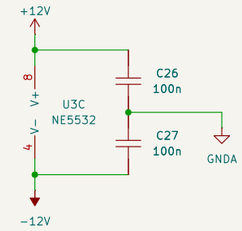
INPUT GAIN AND ADC PAD

The gain pot (10k dual, audio taper) is on the control board. One shielded 3.5 mm stereo cable per channel connects it to J3 / J4.
Pad 5k6 / 1k brings the hot node down to codec level and limits voltage and current at the Daisy pin. No bias: the Seed has its own input capacitor.

LEFT channel: gain pot, x11 (+20.8 dB), pad to the Daisy



U3 supply and decoupling



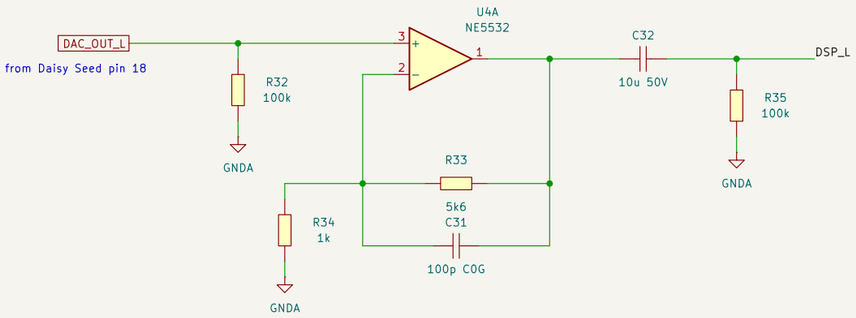
RIGHT channel: gain pot, x11 (+20.8 dB), pad to the Daisy

Sheet: /Input gain and ADC pad/		
File: input-gain.kicad_sch		
Title: Plug & Roll — Main board — Input gain and ADC pad		
Size: A3	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 5/6

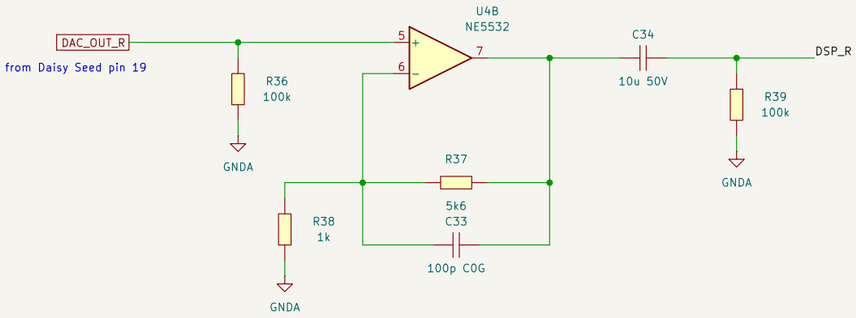
DAC GAIN AND GLOBAL BYPASS

The Daisy output (1.0 Vrms at 0 dBFS) is amplified to the level of the hot node, so the effect path and the analog bypass path reach the bypass relay at the same level. No bias: the Seed has its own output capacitor.

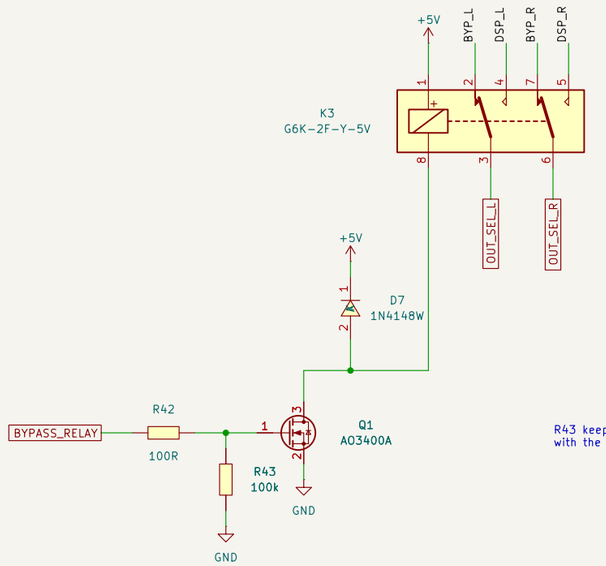
LEFT channel: DAC gain x6.6 (+16.4 dB) = 1 + 5k6 / 1k



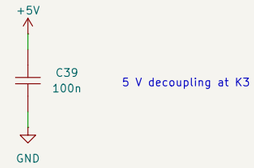
RIGHT channel: DAC gain x6.6 (+16.4 dB) = 1 + 5k6 / 1k



BYPASS RELAY. Drawn released (no coil current) = analog bypass. BYPASS_RELAY high = effect path.

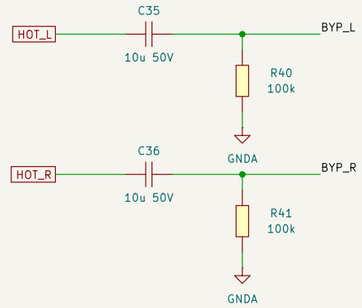


R43 keeps the transistor off until the Daisy drives the line: at power-on, in reset and with the Seed removed the relay stays released = bypass. D7 absorbs the coil's switch-off spike.

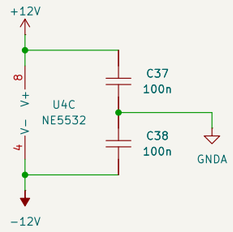


5 V decoupling at K3

Analog bypass signal from the input section



U4 supply and decoupling



Sheet: /DAC gain and bypass/		
File: output-dac.kicad_sch		
Title: Plug & Roll – Main board – DAC gain and global bypass		
Size: A3	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 6/6